UNIVERSITY NAME

Rigid Body Mechanics — Final Exam

Year: 2015

Exercise 1:

1. Show that the velocity vector field of a rigid body defines a torsor.
2. Show that the angular momentum is an antisymmetric vector field.
3. Show the following formula: for any point A belonging to the rigid body S of mass m and center of mass G,

$$L_A(S/R_0) = \Pi_A(S) \cdot \hat{\Omega}(S/R_0) + m \cdot \overrightarrow{AG} \times \overrightarrow{V}(A/R_0).$$

4. (a) Write the theorem of angular momentum about a point A that is mobile in R_0.
   (b) Prove this theorem.

Answer Area

1.

2.

3.

4. (a)
   (b)

Exercise 2:

A physical pendulum S, consisting of a homogeneous rod of length B_a and negligible mass, welded to a homogeneous ball of mass m, radius a, and center of mass G, is in motion relative to a fixed Galilean reference frame $R_0(X_0, Y_0, Z_0)$ with basis $(\bar{x}_0, \bar{y}_0, \bar{z}_0)$.

The system is moving in the vertical plane X_0OY_0 (with $G \in X_0OY_0$) and A is a mobile point. The motion is frictionless at A. R denotes the reaction force exerted on the system at point A.

The position of point A on the axis OY_0 is given by the coordinate y (with $\overrightarrow{OA} = y\overrightarrow{y_0}$) and the position of the system relative to the vertical is defined by the angle θ. The system is displaced from its equilibrium position and then released without initial velocity.

1. Calculate the inertia matrix of the ball at point G.

2. Determine the velocity vector of point G in R_0.

3. By applying the dynamic resultant theorem, show that y and θ are related by the following relation: $y = -6a\dot{\theta}\cos\theta$. Deduce that the kinetic resultant of the system is given by:
$$P(S/R_0) = -6ma\dot{\theta}\sin\theta\overrightarrow{x_0}$$

4. By applying the kinetic energy theorem to the solid, determine the first integral of energy and deduce the differential equation of motion of the solid.

5. Retrieve the differential equation of motion of the solid by applying the angular momentum theorem to the solid, at point A, in the reference frame R_0.

6. Determine the period of small oscillations (keeping only first-order terms in θ and $\dot{\theta}$).

Answer Area

1.
2.
3.
4.
5.
6.